ATM SECURITY SYSTEM

PROJECT REPORT

Submitted by

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Submitted in partial fulfillment of the requirements for award of the degree of

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COLLEGE,MANKARA,PALAKKAD

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CERTIFICATE

This is to certify that the project work entitled

ATM SECURITY SYSTEM

Is the bona fide record of work done by

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For the award for the Degree of BACHELOR OF COMPUTER SCIENCE ,Calicut University Submitted for the viva-voice examination held on

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DECLARATION

We hereby declare that the project entitled "ATM SECURITY SYSTEM" submitted to UNIVERSITY OF CALICUT in partial fulfilment of the requirement for the award of degree in BACHELOR OF COMPUTER SCIENCE is a record of the original work done by me under the guidance ofMr.RAMSHEEL.M , Assistant professor of computer science , during the period of study in SADANAM KUMARAN COLLEGE , MANKARA ,PALAKKAD.

PLACE : PATHIRIPALA

Date :

Signature of the students

AAKASH.V.DEVAN

ATM SECURITY

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INTRODUCTION

1.INTRODUCTION

Machine learning (ML) is the scientific study of algorithms and statistical models that computer systems use to perform a specific task without using explicit instructions, relying on patterns and inference instead. It is seen as a subset of artificial intelligence. Machine learning algorithms build a mathematical model based on sample data, known as "training data", in order to make predictions or decisions without being explicitly programmed to perform the task. Machine learning is closely related to computational statistics, which focuses on making predictions using computers. The study of mathematical optimization delivers methods, theory and application domains to the field of machine learning. "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E.” This is Alan Turing’s definition of machine learning. Deep learning is a class of machine learning algorithms that utilizes a hierarchical level of artificial neural networks to carry out the process of machine learning.

The artificial neural networks are built like the human brain, with neuron nodes connected together like a web. While traditional programs build analysis with data in a linear way, the hierarchical function of deep learning systems enables machines to process data with a nonlinear approach. The word "deep" in "deep learning" refers to the number of layers through which the data is transformed. More precisely, deep learning systems have a substantial credit assignment path (CAP) depth. The CAP is the chain of transformations from input to output. CAPs describe potentially causal connections between input and output. For afeedforward neural network, the depth of the CAPs is that of the network and is the number of hidden layers plus one (as the output layer is also parameterized).

For recurrent neural networks, in which a signal may propagate through a layer more than once, the CAP depth is potentially unlimited. Deep learning architectures such as deep neural networks, deep belief networks, recurrent neural networks and convolutional neural networks have been applied to fields includingcomputer vision, speech recognition, natural language processing, audio recognition, social network filtering, machine translation, bioinformatics, drug design, medical image analysis, material inspection and board game programs, where they have produced results comparable to and in some cases superior to human experts.

Prior to the wide-spread adoption of deep learning systems, computer vision researchers developed a variety of creative and moderately successful methodologies for the automated analysis of helmets based on the raw pixel data from images. Initial approaches for classification required a domain expert to identify meaningful features for the desired classification (such as the unique characteristics of helmet and mask), design an algorithm to extract these features from the image, and compare individual differences using a statistical analysis.

Image processing is the process of transforming an image into a digital form and performing certain operations to get some useful information from it. The image processing system usually treats all images as 2D signals when applying certain predetermined signal processing methods.

There are five main types of image processing:

- Visualization - Find objects that are not visible in the image
- Recognition - Distinguish or detect objects in the image
- Sharpening and restoration - Create an enhanced image from the original image
- Pattern recognition - Measure the various patterns around the objects in the image
- Retrieval - Browse and search images from a large database of digital images that are similar to the original image

One of the most common applications of image processing that we use today is face detection. It follows deep learning algorithms where the machine is first trained with the specific features of human faces, such as the shape of the face, the distance between the eyes, etc. After teaching the machine these human face features, it will start to accept all objects in an image that resemble a human face. Face detection is a vital tool used in security, biometrics and even filters available on most social media apps these days.

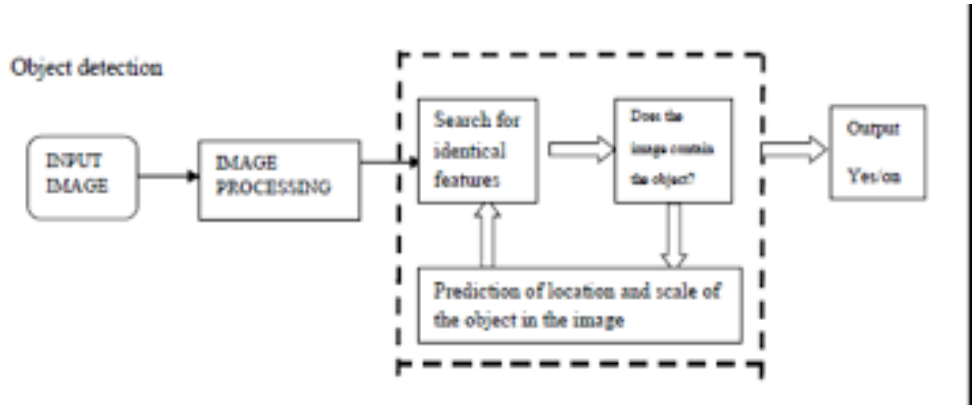
2.ABSTRACT

The automatic teller machine (ATM) plays an important role in the modern economical activity. It provides a fast and convenient way to process economical transactions between banks and their customers. Unfortunately, it also provides a convenient way for criminals to get illegal money .Security Facilities are very important in the place of financial systems. Along with convenience, customers are facing many security problems in ATMs (Automated Teller Machine). As a part of security, surveillance cameras are fixed in each ATM which is monitored by human.

The basic idea behind the project is to develop a system which is automated and provides the most advanced security features in places like ATM. The project aims to give access to ATM's for only those who follow the rules of the ATM usage that is preventing access to ATM for those wearing helmet. CNN (Convolutional Neural Network) Algorithm is used for identification like wearing helmet and mask identifications. Generally video surveillance system at ATM Centre captures every activities of user, but many times any fraudster does fraud wearing Helmet or Mask it is difficult to investigate and find them because of occulted face by helmet or mask. ATM visitors who have wore helmets make it difficult to identify the person if any abnormal activity happens or violate the rules through the surveillance camera. Due to this problem users are not allowed to wear helmet in ATMs. For improving security in ATMs, an automated helmet and mask detection for detecting the helmet. Helmet detection using Tensor Flow increases the safety of ATM users and also increases the safety to the machine itself.With help of above-mentioned attributes, law enforcement department authorities can guess the person involved in crime scene. And those suspected person images are captured, who are carrying weapons in an ATM, sent by mail to the authorized person respectively.

3.WORK FLOW

A workflow diagram (also known as a workflow) provides a graphic overview of the business process. Using standardized symbols and shapes, the workflow shows step by step how your work is completed from start to finish. It also shows who is responsible for work at what point in the process. Designing a workflow involves first conducting a thorough workflow analysis, which can expose potential weaknesses. A workflow analysis can help you define, standardize and identify critical areas of your process.



The above diagram shows how this pipeline generated numerical features and feed it into a machine learning algorithm.

3.1 DATASET

This dataset contains 764 images of 2 distinct classes for the objective of helmet detection.

Bounding box annotations are provided in the PASCAL VOC format

The classes are:

- With helmet
- Without helmet

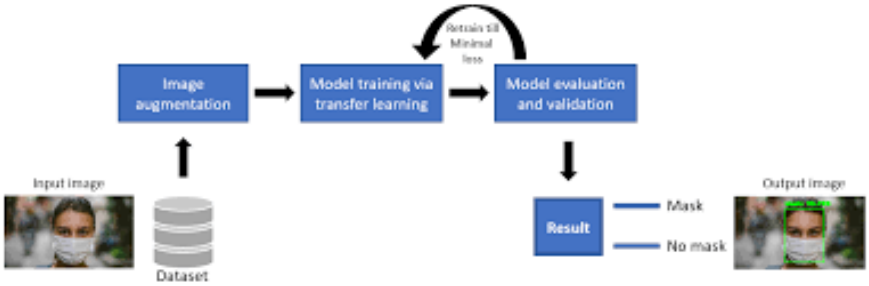
Masks play a crucial role in protecting the health of individuals against respiratory diseases, as is one of the few precautions available for COVID-19 in the absence of immunization. With this dataset, it is possible to create a model to detect people wearing masks, not wearing them, or wearing masks improperly.

This dataset contains 2994 images belonging to the 2 classes, as well as their bounding boxes in the PASCAL VOC format.

The classes are:

- With mask
- Without mask

- **Training data.** This type of data builds up the machine learning algorithm. The data scientist feeds the algorithm input data, which corresponds to an expected output. The model evaluates the data repeatedly to learn more about the data's behavior and then adjusts itself to serve its intended purpose.
- **Validation data.** During training, validation data infuses new data into the model that it hasn't evaluated before. Validation data provides the first test against unseen data, allowing data scientists to evaluate how well the model makes predictions based on the new data. Not all data scientists use validation data, but it can provide some helpful information to optimize hyperparameters, which influence how the model assesses data.
- **Test data.** After the model is built, testing data once again validates that it can make accurate predictions. If training and validation data include labels to monitor performance metrics of the model, the testing data should be unlabeled. Test data provides a final, real-world check of an unseen dataset to confirm that the ML algorithm was trained effectively.



SYSTEM ANALYSIS

4.SYSTEM ANALYSIS

4.1 Existing system

All ATMs are provided with CCTV to know who is operating the ATMS and what they are doing inside the ATMS. When there are misuse of ATMs by fraudsters the bank wants to know the identity of the person using the ATMs so that they can trace the person and find the culprit if there is misuse of card or any other fradulent transaction is taking place. If the person using the ATM is using helmet then his identity will not be clear and it will be difficult to locate the person who used the ATM. Hence the rule of helmets not allowed in ATMs.

The existing methods are not suitable to identify each bizzare situation occurs inside the ATM counters. For safety reasons, each ATM is with the surveillance system to record customer's face and activities . The existing CCTV system will only identify the abnormal event through video clips .However when criminals use the ATM to withdraw illegal money , they usually hide their faces with something (e.g. safety helmets) to avoid the surveillance system to record their face informations. That will make the system decrease its efficiency.

4.1.1 Limitations of Existing System

- CCTV just stores only the information.
- Alert messages will not be sent to the concerened peoples(bank office or nearby police stations).

4.2 PROPOSED SYSTEM

The proposed system will provide much security to the ATM machine than the existing system. This architecture contains a camera that detects the presence of helmet when anyone wearing a helmet came infront of the machine.The user cannot proceed with the next step unless he is ready to remove the same.Also when the entered pin doesnot match with the customer’s debit card’s pin then he/she is warned to remove the mask in case a mask is detected and capture the user’s face .

The proposed method can be applied to the surveillance systems of ATMs and banks, and it can provide the early warning to save-guards when any "customer" tries to avoid his/her face information from surveillance, such as withdrawing money with the safety helmet. That will make the surveillance system more useful. A real image is used to see the performance of proposed method.

SYSTEM STUDY

5.SYSTEM STUDY

5.1 FEASIBILITY STUDY

Feasibility study is a process that identifies, describes and evaluates proposed system and selects the best system.It is an evaluation of a proposal designed to determine the difficulty in carrying out a designated task. Generally, a feasibility study precedes technical development and project implementation. In other words, a feasibility study is an evaluation or analysis of the potential impact of a proposed project.

Feasibility Study is performed to choose the system that meets the performance requirements at least cost. The most essential tasks performed by a Feasibility Study are the identification and description of candidate systems, the evaluation of the candidate systems and the selection of the best of the candidate systems. The best system means the system that meet performance requirements at the least cost. The most difficult part of a Feasibility Study is the identification of the candidate systems and the evaluation of their performances and costs. So it is worth solving the problem.. During the study, the problem definition is solved and all aspects of problem to be included in the system are determined. size of project, cost of benefits is also estimated with greater accuracy. a good feasibility study will show the strength and defects before the project is planned or budgeted.

5.2 ECONOMICAL FEASIBILITY

The purpose of the economic feasibility assessment is to determine the positive economic benefits to the organization that the proposed system will provide. It includes quantification and identification. It identifies the financial benefits and cost associated with the development of the system. Economic feasibility is often known as the cost benefit analysis. To carry out an economic feasibility study it is necessary to estimate actual money value against activities needed for implementing the system. While implementing our system we can ensure that the cost of prospective new venture will ultimately be profitable to the people. So we can say it is financially feasible.

5.3 TECHNICAL FEASIBILITY

It determines whether the technology is needed for the proposed system is available and how it can be integrated with the government. Technical evaluation must assess whether the user have technical expert to understand and use the new system. This assessment is based on an outline design of system requirements, to determine whether proposal is technically feasible. It is the evaluation of the hardware and software and how it meets the need of the proposed system. The technology used can be developed with the current equipments and has the technical capacity to hold data required by the old system.

This technology supports the modern trends of technology. Easily accessible more secure technologies.

5.4 OPERATIONAL FEASIBILITY

Operational feasibility is the measure of how well a proposed system solves the problems, and takes advantage of the opportunities identified during scope definition and how it satisfies the requirements identified in the requirement analysis phase of system development. it focuses on the degree to which the proposed development project fits in with the existing environment and objectives it regards to development schedule. Operational feasibility focus on human, organizational and political.

The proposed system can easily implement, as this is based on Python The resources that are required to implement/install these are available. So the project is operationally feasible.

5.5 BEHAVIOURAL FEASIBILITY

It includes how strong the reaction of people will be towards the development of new system that involves computers use in their daily world. So, the resistant to change is identified. Behavioural feasibility refers to the system which will be designed to process data and produce the desired

outputs. However, if the data input for the system is not readily available or collectable, then the system may not be successful.

Through our system we introduce an efficient way to apply for store childern's vaccinated data. This can be easily accepted by the government, because they carry a good knowledge of computerized systems. so, our project is feasible in behaviour manner too.

SYSTEM SPECIFICATION

6.SYSTEM SPECIFICATION

The selection of hardware is very important in the existence and proper working of any software

6.1 SOFTWARE REQUIREMENTS

The software configuration say about the softwares and packages used to design the system. The following specifications includes the software and packages used by the programmer.

- Platform: anaconda navigator
- Front end tool :tkinter
- Back end tool : python
- Operating system :windows 10
- Server :

6.2 HARDWARE REQUIREMENTS

PROCESSOR : INTEL CORE

RAM : 4 GB

HARD DISK : 1 TB

CODE DESIGN

7.CODE DESIGN

The coding step transfers a detailed design of representation of software into a programming language realization. All the software engineering steps that have been presented up to this step are directed towards a final objective. Coding is a process that transforms design into a programming language. The translation process begins when a compiler accepts the source code as input and produces machine dependent object code as output. Compiler output is further translated into machine code. Language characteristics have an impact on the quality and efficiency of translation. Building the application refers to the coding step in software engineering process.

Wehave done this project by using Python as the developing language. Since prototype model it doesn't require any database. Also, Here Anaconda Navigator is as an Integrated Development Environment .

7.1 Python

Python is an interpreted high-level language general purpose programming language. Its design philosophy emphasizes code readability with its use of significant identification. its language constructs as well as it object-oriented approach aim to help programmers write clear, logical code for small and large-scale projects.

Python is dynamically-typed and garbage-collected. It supports multiple programming paradigms, including structured (particularly procedural), object-oriented and functional programming. It is often described as a "batteries included" language due to its comprehensive standard library.

Guido van Rossum began working on python in the late 1980s, as a successor to the ABC programming language, and first released it in 1991 as python 0.9.0. python 2.0 was released in 2000 and introduced new features, such as listcomprehensions and a cycle detecting garbage collection system (in addition to reference coding). Python 3.0 was released in 2008 and was a major revision of the language that is not completely backward-compatible. Python 2 was discontinued with version 2.7.18 in 2020. Python consistently ranks as one of the most popular programming languages.

FEATURES OF PYTHON:

- Easy to code
- Object-Oriented language
- Free and open source
- GUI programming support
- High level language
- Extensible feature
- Portable
- Interpreted
- Integrated language
- Large standard library
- Dynamically typed language

7.1.1 TKinter

Tkinter is the Python interface to Tk, which is the GUI toolkit for Tcl/Tk.

Tcl (pronounced as tickle) is a scripting language often used in testing, prototyping, and GUI development. Tk is an open-source, cross-platform widget toolkit used by many different programming languages to build GUI programs.

Python implements the Tkinter as a module. Tkinter is a wrapper of C extensions that use Tcl/Tk libraries.

Tkinter allows you to develop desktop applications. It’s a very good tool for GUI programming in Python.

Tkinter is a good choice because of the following reasons:

- Easy to learn.
- Use very little code to make a functional desktop application.
- Layered design.
- Portable across all operating systems including Windows, macOS, and Linux.
- Pre-installed with the standard Python library.

7.1.2 tkinter.ttk

The tkinter.ttk module provides access to the Tk themed widget set, introduced in Tk 8.5. If Python has not been compiled against Tk 8.5, this module can still be accessed if *Tile* has been installed. The former method using Tk 8.5 provides additional benefits including anti-aliased font rendering under X11 and window transparency (requiring a composition window manager on X11).

The basic idea for tkinter.ttk is to separate, to the extent possible, the code implementing a widget's behavior from the code implementing its appearance.

Ttk comes with 18 widgets, twelve of which already existed in
tkinter: Button, Checkbutton, Entry, Frame, Label, LabelFrame, Menubutton, PanedWindow, Radiobutton, Scale, Scrollbar, and Spinbox. The other six are
new: Combobox, Notebook, Progressbar, Separator, Sizegrip and Treeview. And all them are subclasses of Widget.

7.1.3 Open CV

OpenCV (Open Source Computer Vision Library) is an open source computer vision and machine learning software library. OpenCV was built to provide a common infrastructure for computer vision applications and to accelerate the use of machine perception in the commercial products. Being a BSD-licensed product, OpenCV makes it easy for businesses to utilize and modify the code.

The library has more than 2500 optimized algorithms, which includes a comprehensive set of both classic and state-of-the-art computer vision and machine learning algorithms. These algorithms can be used to detect and recognize faces, identify objects, classify human actions in videos, track camera movements, track moving objects, extract 3D models of objects, produce 3D point clouds from stereo cameras, stitch images together to produce a high resolution image of an entire scene, find similar images from an image database, remove red eyes from images taken using flash, follow eye movements, recognize scenery and establish markers to overlay it with augmented reality, etc. OpenCV has more than 47 thousand people of user community and estimated number of downloads exceeding 18 million. The library is used extensively in companies, research groups and by governmental bodies.

Along with well-established companies like Google, Yahoo, Microsoft, Intel, IBM, Sony, Honda, Toyota that employ the library, there are many startups such as Applied Minds, VideoSurf, and Zeitera, that make extensive use of OpenCV. OpenCV's deployed uses span the range from stitching streetview images together, detecting intrusions in surveillance video in Israel, monitoring mine equipment in China, helping robots navigate and pick up objects at Willow Garage, detection of swimming pool drowning accidents in Europe, running interactive art in Spain and New York, checking runways for debris in Turkey, inspecting labels on products in factories around the world on to rapid face detection in Japan.

It has C++, Python, Java and MATLAB interfaces and supports Windows, Linux, Android and Mac OS. OpenCV leans mostly towards real-time vision applications and takes advantage of MMX and SSE instructions when available. A full-featured CUDAand OpenCL interfaces are being actively developed right now. There are over 500 algorithms and about 10 times as many functions that compose or support those algorithms. OpenCV is written natively in C++ and has a templated interface that works seamlessly with STL containers.

7.1.4 NumPy

NumPy is a Python library used for working with arrays.

It also has functions for working in domain of linear algebra, fourier transform, and matrices.

NumPy was created in 2005 by Travis Oliphant. It is an open source project and you can use it freely.

NumPy stands for Numerical Python.

7.1.5 TensorFlow

TensorFlow is an end-to-end open source platform for machine learning. It has a comprehensive, flexible ecosystem of tools, libraries and community resources that lets researchers push the state-of-the-art in ML and developers easily build and deploy ML powered applications.

TensorFlow offers multiple levels of abstraction so you can choose the right one for your needs. Build and train models by using the high-level Keras API, which makes getting started with TensorFlow and machine learning easy.

If you need more flexibility, eager execution allows for immediate iteration and intuitive debugging. For large ML training tasks, use the Distribution Strategy API for distributed training on different hardware configurations without changing the model definition.

7.2 Anaconda Navigator

Anaconda is the distribution of the python and R programming languages for scientific computing(data science, machine learning applications, large-scale data processing, predictive analysis, etc.) that aims to simplify package management and deployment. The distribution includes data-science

packages suitable for Windows, Linux, and macOS. It is developed and maintained by Anaconda, Inc., which was founded by Peter Wang and Travis Oliphant in 2012. As an Anaconda, Inc. product, it is also known as Anaconda Distribution or Anaconda Individual Edition, while other products from the company are Anaconda Team Edition and Anaconda Enterprise Edition, both of which are not free.

Package versions in Anaconda are managed by the package management system conda. This package manager was spun out as a separate open-source package as it ended up being useful on its own and for things other than Python. There is also a small, bootstrap version of Anaconda called Miniconda, which includes only conda, Python, the packages they depend on, and a small number of other packages.

Anaconda is freely available, open source distribution of python and R programming languages which is used for scientific computations. If you are doing any machine learning or deep learning project then this is the best place for you. It consists of many softwares which will help you to build your machine learning project and deep learning project. these softwares have great graphical user interface and these will make your work easy to do. you can also use it to run your python script.

In order to run, many scientific packages depend on specific versions of other packages. Data scientists often use multiple versions of many packages and use multiple environments to separate these different versions.The command-line program conda is both a package manager and an environment manager. This helps data scientists ensure that each version of each package has all the dependencies it requires and works correctly.Navigator is an easy, point-and-click way to work with packages and environments without needing to type conda commands in a terminal window. You can use it to find the packages you want, install them in an environment, run the packages, and update them – all inside Navigator.

7.3 MACHINE LEARNING (ML)

ML is the study of computer algorithms that can improve automatically through experience and by the use of data. It is seen as a part of artificial intelligence. Machine learning algorithms build a model based on sample data, known as training data, in order to make predictions or decisions without being explicitly programmed to do so.[1] Machine learning algorithms are used in a wide variety of applications, such as in medicine, email filtering, speech recognition, and computer vision, where it is difficult or unfeasible to develop conventional algorithms to perform the needed tasks.

A subset of machine learning is closely related to computational statistics, which focuses on making predictions using computers; but not all machine learning is statistical learning. The study of mathematical optimization ,delivers methods, theory and application domains to the field of machine learning. The term *machine learning* was coined in 1959 by Arthur Samuel, an American IBMer and pioneer in the field of computer gaming and artificial intelligence. Also the synonym *self-teaching computers* was used in this time period. A representative book of the machine learning research during the 1960s was the Nilsson's book on Learning Machines, dealing mostly with machine learning for pattern classification. Interest related to pattern recognition continued into the 1970s, as described by Duda and Hart in 1973 In 1981 a report was given on using teaching strategies so that a neuralnetwork learns to recognize 40 characters (26 letters, 10 digits, and 4 special symbols) from a computer terminal.

Tom M.Michel provided a widely quoted, more formal definition of the algorithms studied in the machine learning field: "A computer program is said to learn from experience *E* with respect to some class of tasks *T* and performance measure *P* if its performance at tasks in *T*, as measured by *P*, improves with experience *E*. This definition of the tasks in which machine learning is concerned offers a fundamentally operational division rather than defining the field in cognitive terms. This follows Alan Turing's proposal in his paper "Computing machinery and intelligence", in which the question "Can machines think?" is replaced with the question "Can machines do what we (as thinking entities) can do?".

Modern day machine learning has two objectives, one is to classify data based on models which have been developed, the other purpose is to make predictions for future outcomes based on these models. A hypothetical algorithm specific to classifying data may use computer vision of moles coupled with supervised learning in order to train it to classify the cancerous moles. A machine learning algorithm for stock trading may inform the trader of future potential predictions.

SYSTM DESIGN

8.SYSTM DESIGN

8.1 INTRODUCTION TO SYSTEM DESIGN

System design deals with transforming the customer requirements, into a form that is implemented using a programming language. It is the phase in which, the detailed design of the system is accomplished and the user oriented performance specification converted into a technical design specification. During this phase, decision about overall architecture was taken. The system organized into subsystem based on the analysis structure. A personal computer with ordinary specification was chosen for the implementation of the design. The major activities include general system design, design of all outputs, input design and the design of the database.

8.2 INPUT DESIGN

Input design is a part of overall design, which requires careful attributes. Input design is a process of converting user-oriented input into computer-based format. This input to the system is provided using .the goal of designing input data is to make entry as easy, logical and free from errors. Computers require necessary data for their functioning, hence input design become an essential part of computer oriented information system. In input design user oriented inputs are converted into computer based formats. Inputs are of two types in a computer system. Input document used in a system input screens. The goal of designing input data is to make data entry easy, logical and free from error as possible. Inputs are raw data that are accepted from the system and are processed to produce the output. The input records have to be validated, organized and accepted by the system before being processed to produce the output. We use some data entry screens for inputting different data. Input screens messages and careful programming can reduce input as errors to some extent. Project report text page | 20 malabar college of commerce and science input design features can ensure the reliability of the system and also determines how efficiently the user can interact with the system. A good input screen is clear and free from irrelevant information. Entered data has to be checked for their accuracy and detection of errors. All the data entry screens should be of interactive nature so that the user can directly input data according to prompt messages. The input design determines whether the user

can interact directly with the computer. Interactive input screens ensure the reliability and accuracy of the system. All the input screens used here have script validations for null checking and data types wherever necessary. All the programs inserting data to the table checks the length of the values in addition to the data type verification. They also prompt the entry of values in mandatory fields. The goal of designing input data is to make it free from logical errors. The input data is also used for easy calculation of necessary functions. Different forms are used for data screens in order to input data into the system. The screen formats have options like add, delete, edit and update with buttons, which is used for easy input and retrieving information. All the fields are validated. if the user enters invalid data, appropriate messages are displayed. Input specifications describe the manner in which data enter the system for processing. The design of the input should have:

• Effectiveness

• Accuracy

• Simplicity

• Ease to use

• Consistency

• Attractiveness

8.3 OUTPUT DESIGN

- Computer output is the most important and direct source of information to the user. Efficient intelligible output design improves the system relationship with the user and helps in decision making.
- One of the most important features of the system for users is the output it produces. Output design should improve the system relationship with the user it produces and helps in decision making. considering the future use of the output required and depending on the nature, they are displayed on the monitor for immediate need or obtaining the hard copy

- The objective of output design is to define the control and formats of all printed, documented, reports and screens that will be produced by the system. Computer output is the most important and direct source of information to the user. For many end users output is the main reason for developing the system and the basis on which they will evaluate the usefulness of the application. Output generally refers to the results that are generated by the system. The output of the system is designed so as to include a number of reports. Reports reflect the output design. objective of output design are:
- Design output to serve the intended purpose.
- Design output to fit the user and choose the right output method.
- Deliver the appropriate quantity of output and provide output on time.

8.4 DFD (Data Flow Diagram)

A DFD, also known as a “bubble chart” has the purpose of clarifying system requirements and identifying major transformations that will become programs in system design. A DFD consist of a series of bubbles joined by lines. The bubbles represent data transformations and the lines represent data flow in the system. A data flow diagram may be used to represent a system or software at any level of abstraction.

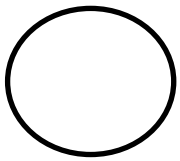
DFD’s can be partitioned into levels that represent increasing functional details. A DFD is a diagram that describes the flow of data and the process that change or transform data throughout a system. It is a structured analysis and design tool that can be used for flow-charting in a place of, or in association with, information oriented and process-oriented system flow charts. When analysts prepare the DFD they specify the user needs at a level of detail that virtually determines the information flow into and out of the required data resources. This network is constructed using a set of symbols that do not imply a physical implementation. The DFD reviews the current physical system, prepares input and output specifications, and specifies the implementation plan etc.

The basic elements of DFD are: -

- Bubble: Used to represent a function.
- Arrows: Used to represent data flow.
- Rectangle: Used to represent external entities.
- Open rectangle: Used to represent data store.

Basic Data Flow Diagram Notations

A data flow is a route, which enables packets of data to travel from one point to another. Data may flow from a source to a processor and from data store or process. An arrow line depicts the flow, withanarrowhead pointing in the direction of flow.



A process represents transformation where incoming data flows are changed into outgoing data flows.

A data store is a repository of data that is to be stored for use by one or more process may be as simple as a buffer or queue or sophisticated as a relationaldatabase.

A source or sink is a person or part of an organization, which enters or receives information from the system, but is outside the contest of data flow model.

Steps to construct Data Flow Diagram:

Four steps are basically used to construct DFD,

 Process should be named a number for easy reference. Each name should be representative of the process.

- The direction of flow is from top to bottom and from left to right.
- When a process is exploded into lower level details they are numbers.
- The name of data stores, sources and destination are written in capital letters

LEVELS OF DFDs

- **Level 0**

The DFD Level 0 is also know as the context diagram of ATM Management System. ATM Management System context diagram presents the main idea to be the basis for the proceeding levels. The basic idea is represented by a single process consisting the main process, users and data.

- **Level 1**

Next to the context diagram is the level 1 data flow diagram. ATM Management System DFD Level 1 is the broadened context terms that consist of several processes derived from the main process. They were also numbers to see that were all part of he single process form ATM Management System DFD Level 0.

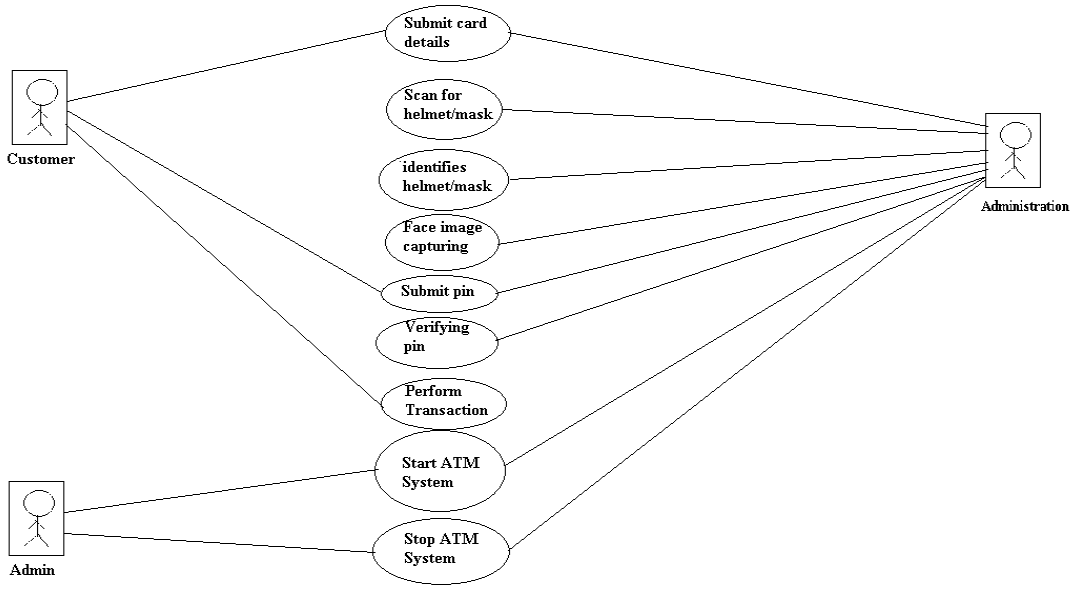
- **Level 2**

After presenting the ATM system DFD levels 0 and 1, next to that is level 2. DFD 2 shows a diagram where we add the data storage into the ATM Management System function. Since everything were all set, all we have to justify is the security of the data that enters and exits the system.

8.5 USE CASE DIAGRAM

A use case diagram is a graphical depiction of a user's possible interactions with a system. A use case diagram shows various use cases and different types of users the system has and will often be accompanied by other types of diagrams as well. The use cases are represented by either circles or ellipses. While a use case itself might drill into a lot of detail about every possibility, a use-case diagram can help provide a higher-level view of the system. It has been said before that "Use case diagrams are the blueprints for your system".

Due to their simplistic nature, use case diagrams can be a good communication tool for stakeholders. The drawings attempt to mimic the real world and provide a view for the stakeholder to understand how the system is going to be designed. Siau and Lee conducted research to determine if there was a valid situation for use case diagrams at all or if they were unnecessary. What was found was that the use case diagrams conveyed the intent of the system in a more simplified manner to stakeholders and that they were "interpreted more completely than class diagrams".



SYSTEM TESTING

9.SYSTEM TESTING

System testing is the stage of implementation, which is aimed at ensuring that the system works accurately and efficiently before live operation commences. For any software that is newly developed, primary importance is given to testing the system .It is the last opportunity for the developer over to the customers. Testing is the process by which a developer will generate a set of test data, which gives maximum probability of finding all types of errors that can occur in the software. Testing is vital to the success of the system. System testing makes a logical assumption that if all the parts of the system are correct, the goal will be successfully achieved. The candidate system is subject to a variety of tests: online response, volume, stress, recovery & security and usability tests. A series of testing are performed for the proposed system before the system is ready for user acceptance testing.

It is the process of exercising or evaluating a system by manual or automatic means to verify that it satisfies the specified requirements or to identify the difference between expected and actual results. The testing activities are aimed at convincing the customer through demonstration and actual use that the software is a solution to the original problem and that both the product and the process that created it are of high quality. It is also used to find and eliminate any residual errors from previous stages and the operational reliability of the system.

9.1 PREPARATION OF TEST DATA

A test set in machine learning is a secondary (or tertiary) data set that is used to test a machine learning program after it has been trained on an initial training data set. The idea is that predictive models always have some sort of unknown capacity that needs to be tested out, as opposed to analyzed from a programming perspective.A test set is also known as a test data set or test data.

9.2 GOALS AND OBJECTIVES

The main purpose of testing an information system is to find the errors and correct them. The scope of system testing should include both manual and computerized operations. System testing is comprehensive evaluation of the programs, manual procedures, computer operations and controls. System testing is the process of checking whether the developed system is working according to the objective and requirement. All testing is to be conducted in accordance to the test conditions specified earlier. This will ensure that the test coverage meets the requirements and that testing is done in a systematic manner.

9.3 TESTING OBJECTIVES:

- Testing is the process of executing a program, with the intent of finding so many errors as possible.
- A good test case is one that has a high probability of finding an as-yet-undiscovered error.
- A successful test is one that uncovers an as-yet-undiscovered error.

So, the main objective is to design tests that systematically uncover different classes of errors using minimum time and effort. Successful testing uncovers errors in software. It also shows that the software functions are working according to specifications. Also, the data collected during testing provides an indication of software reliability and software quality.

9.4 TESTING TECHNIQUES

Testing is vital to the success of the system. System testing makes a logical assumption that if all parts of the system are subject to variety of tests on-line response, volume, stress, recovery and security and usability tests. Once the system performs flawlessly on artificial data, we switch to 'Live Data' or real data taken from the organization. A system is generally tested in a hierarchical fashion starting at the bottom and working up. First each program is tested; next a series of modules is tested; then each individual program with all its modules; finally the entire

system consisting of a series of programs is tested. In this way, problems at the module level can be corrected before programs are tested and problems at the program level can be corrected before the entire system is used. A series of tests are performed before the system is ready for user acceptance testing.

9.5 SOFTWARE TESTING

Software testing is a critical element of software quality assurance and represents the ultimate reviews of specification, design and coding. Testing is presents an interesting anomaly for the software. Testing is vital to the success of the system. Errors can be injected at any stage during development. System testing makes a logical assumption that if all the parts of the system are correct, the goal will be successfully achieved. During testing, the program to be tested is executed with set of test data and the output of the program for the test data is evaluated to determine if the program is performing as expected. A series of testing are performed for the proposed system before the system is ready for user acceptance testing. The testing steps are:

- Unit Testing
- Integration Testing
- Validation testing
- Output Testing
- Acceptance Testing

9.5.1 Unit Testing

Is a software testing technique by means of which individual units of software i.e. group of computer program modules, usage procedures and operating procedures are tested to determine whether they are suitable for use or not. It is a testing method using which every independent modules are tested to determine if there are any issue by the developer himself. It is correlated with functional correctness of the independent modules.Unit Testing of software product is carried out during the development of an application. An individual component may be either an individual function or a procedure. Unit Testing is typically performed by the developer.

9.5.2 Integration Testing

Integration testing is a type of testing where individual software modules are logically integrated and tested as a group. This testing is done to test that individual modules work as expected when they are combined. Each individual module (units) is first tested in isolation. Once unit testing has been performed, these units are integrated. The integration testing is then performed to validate that the modules work as expected when combined. Integration testing is not necessarily performed at the end of software development. Instead, it is performed throughout the development life cycle as the individual modules get developed.

9.5.3 Validation Testing

Validation testing can be defined in many ways, but a simple definition is that validation succeeds when the software functions in manner that is reasonably expected by the customer. Software validation is achieved through a series of black box tests that demonstrate conformity with requirement. After validation test has been conducted, one of two conditions exists.

- The function or performance characteristics confirm to specifications and are accepted
- A validation from specification is uncovered and a deficiency created.

Deviation or errors discovered at this step in this project is corrected prior to completion of the project with the help of the user by negotiating to establish a method for resolving deficiencies.

Thus the proposed system under consideration has been tested by using validation testing and found to be working satisfactorily.

9.5.4 Output Testing

After performing the validation testing, the next step is output testing of the proposed system, since no system could be useful if it does not produce the required output in the specific format. The output generator or displayed by the system under consideration is tested by asking the users about the format required by them. Here the output is considered in two ways: One is on screen and the other is printed format. The output format on the screen is found to be correct as the format was designed in the system design phase according to the user needs. As far as hardcopies are considered it goes in terms with the user requirement. Hence output testing does not result any correction in the system.

9.5.5 Acceptance Testing

User acceptance of the system is key factor for the success of any system. The system under consideration is tested for user acceptance by constantly keeping in touch with prospective system and user at the time of developing and making changes whenever required.

9.6 White Box Testing

It is the testing method in which internal codes & structure of the software is known to the tester. The tester chooses inputs and exercises the test by giving inputs to the system through the codes and determines the appropriate outputs. The main focus of ***WBT*** is on strengthening the security and on improving the design and usability of the software.

For this, the tester is supposed to have the knowledge of programming languages and implementation of codes. The tester should be aware of the security concerns of the project and his job is to prevent the attacks from users & hackers which may inject malicious code into the software either intentionally or unintentionally. The White Box Testing is like the working of a car mechanic who knows all the internals of the car and checks why it is not working or giving less mileage etc.

White Box Testing is also known as Glass Box Testing, Clear Box Testing, Open Box Testing, Logic Driven Testing, Path Driven Testing or Structural Testing**.**

Definition by ISTQB

- **White-Box Testing***:* Testing based on an analysis of the internal structure of the component or system.
- **White-Box Test Design Technique**: Procedure to derive and/or select test cases based on an analysis of the internal structure of a component or system.

9.7 Black Box testing

Black box testing is a technique of software testing which examines the functionality of software without peering into its internal structure or coding. The primary source of black box testing is a specification of requirements that is stated by the customer.

In this method, tester selects a function and gives input value to examine its functionality, and checks whether the function is giving expected output or not. If the function produces correct output, then it is passed in testing, otherwise failed. The test team reports the result to the development team and then tests the next function. After completing testing of all functions if there are severe problems, then it is given back to the development team for correction.

ISTQBDefinition

- **black box testing:** Testing, either functional or non-functional, without reference to the internal structure of the component or system.
- **black box test design technique:** Procedure to derive and/or select test cases based on an analysis of the specification, either functional or non-functional, of a component or system without reference to its internal structure.

IMPLEMENTATION

10.IMPLEMENTATION

System implementation is the final phase i.e., putting the utility into action. Implementation is the state in the project where theoretical design turned into working system. The most crucial stage is achieving a new successful system and giving confidence in new system that it will work efficiently and effectively. The system is implemented only after thorough checking is done and if it is found working in according to the specifications.

It involves careful planning, investigation of the current system and constraints on implementation, design of methods to achieve. Two checking is done and if it is found working according to the specification, major task of preparing the implementation are educating, training the users.

The implementation process begins with preparing a plan for the implementation of the system. According to this plan, the activities are to be carried out, discussions made regarding the equipment and resources and the additional equipment has to be acquired to implement the new system. The most important in implementation stage is, gaining the users confidence that the new system will work and be effective. The system can be implemented only after through testing is done. This method also offers the greatest security since the existing system can take over if the errors are found or inability to handle certain type of transactions while using the new system.

10.1 IMPLEMENTATION METHODS

There are several methods for handling the implementation and consists for changing from the old to the new computerized system. The most secure method for conversion from the old system is to run the old and new system in parallel .In this approach; a person may operate in the manual processing system as well as start operating the new computerized system.

Another commonly used method is a direct cut over the existing manual system to the computerized system. The change may be within a week or a day. This strategy requires planning. A working version of the system can also be implemented in one part of the

organization and the changes can be made as and when required, but this method is less preference due to the loss of entire system. After the system is Implementation, a review should be conducted to determine whether the system is meeting expecting where improvements are needed.

Implementation is the process of bringing a developed system into operational use and turning it over to the user. Implementation includes all those activities that take place to convert from old system to new. At this stage the theoretical design is turned into a working system. The crucial stage in achieving a successful new system and giving confidence on the system for the users that will work efficiently and effectively.

- The implementation stage involves the following tasks:
- Careful planning
- Investigation of the current system and its constraints
- Design of methods to achieve the changeover
- Training of staff in the overall procedures
- Evaluation of changeover

10.2 IMPLEMENTATION PLAN

Implementation plan includes a description of all activities that must occur to implement the new system and to put into operation. It defines the personal responsible for the activities and prepares a time chart for Implementation the system. The Implementation plan should anticipate possible problems and must be able to deal with them. The usual problem may be missing documents, missed data formats between current and new files, errors in data translation, missing data etc.

Training plats are an important element of the implementation plan. Their purpose is to ensure that all the persons who are associated with the computer based business system possess the necessary knowledge and skills.

Operator training

Operator training is completed in conjunction with its installation and checkout. Operators must become familiar with operational requirements of the new systems. Well prepared manual provide a ready reference to specific duties and step by step operation instruction.

User Training

After the system is implemented successfully, training of the user is one of the most important subtasks of the developer. Even well designed and technically elegant systems can succeed or fail, because of the way they are operated and used. For this purpose user manuals are prepared and handled over to the user to operate the developed system.

Thus the users are trained to operate the developed system. Both the hardware and software securities are made to run the developed systems successfully in future. In order to put new application system into use, preparation of user and system documentation, conducting user training with demo, test run for some period to ensure smooth switching over the system are to be prepared.

Documentation

The documentation involves collecting, organizing, and maintaining complete record of programs. The documentation deals with the system department with maximum clarity. Each and every process is explained in detail. The various table used by the system with field details are provided. The system uses various kinds of forms to produce well-structured screen formats. These forms are also documented. The output generated by the system constitutes another part. Documentation of the software provides the following:

Comments

Comments are very useful in documenting a program. It is used to explain logic of the program. It should be used to improve the quality and understanding of the program. It should not be redundant, incorrect or incomplete.

System Manuals

A good software system must contain standard system manuals. In this the statement is clearly defined, specifies description, detailed flowcharts, and specimen of all input forms and printed outputs.

Operation Manual

A good software package is supported with a good operation manual to ensure the smooth running of the program.

The operation manual must contain the following information:

- Setup and operational details of each program.
- Loading and unloading procedures.
- Starting, running, and terminating procedures.
- List of error conditions with explanations

SYSTEM MAINTANENCE

11. 1 Future Enhancement

Further enhancement is needed to develop this project. In this project we can add modules and multi-layered security system. We can make the security measures in a better way by providing backup and recovery by the admin. Modifications and improvements can be done to make the project more users friendly. Also give more security for further improvisation we can add an alert call after closing the bank. On the other hand, the conventional state of the art image processing algorithms available for occluded and covered face detection, human abnormal behavior analysis and illegal object detection,abnormal gestures, and crime devices.We can also add Biometrics technology.Instead of inserting a card into an ATM machine ,biometrics technology like fingerprint allow customer to use their fingerprint to initiate any banking transaction.

CONCLUSION

10.CONCLUSION

The paper proposes a method for detecting the wearing of safety helmets and mask inside the ATM counter by using the image processing technology. A dataset of around 3500 images with and without helmet and mask is built and divided into four parts(with helmet,without helmet,with mask and without mask) to train and test the model.Tensorflow framework is choosen to train the images.The presented method offers an alternative solution to detect the safety helmets and mask worn by the customers inside the ATM counter.Thus our project enhance the security of ATM. If anyone enteringthe ATM by wearing a helmet cannot withdraw money unless he is ready to remove it and only then the further transaction takes place.

ANNEXURE

10.ANNEXURE

13.1 Project Coding

```python


# This will import all the widgets
# and modules which are available in
# tkinter and ttk module
from tkinter import *
from tkinter.ttk import *
import os
import cv2
import numpy as np


# creates a Tk() objec
win = Tk()

# sets the geometry of main
# root window
win.geometry("400x400")


def warning():
    #Create an instance of Tkinter frame
    win= Tk()

    #Set the geometry of Tkinter frame
    win.geometry("750x250")

```

```python
    def display_text():
        global entry
        string= entry.get()
        label.configure(text="PLEASE REMOVE YOUR HELMET")

    #Initialize a Label to display the User Input
    label=Label(win, text="", font=("Courier 22 bold"))
    label.configure(text="PLEASE REMOVE YOUR HELMET")
    label.pack()
    ttk.Button(win, text= "Okay",width= 20, command= a).pack(pady=20)
# function to open a new window
# on a button click


def a():

    import os
    import cv2
    import numpy as np
    from tensorflow.keras.models import model_from_json
    from tensorflow.keras.preprocessing import image


    i,j,k=0,0,0

    #load model
    model = model_from_json(open("model.json", "r").read())
    #load weights
    model.load_weights('model.h5')


    face_haar_cascade = cv2.CascadeClassifier('haarcascade_frontalface_default.xml')
```

```python
cap=cv2.VideoCapture(0)

while True:
    ret,test_img=cap.read()# captures frame and returns boolean value and captured image
    if not ret:
        continue
    gray_img= cv2.cvtColor(test_img, cv2.COLOR_BGR2GRAY)

    faces_detected = face_haar_cascade.detectMultiScale(gray_img, 1.32, 5)


    for (x,y,w,h) in faces_detected:
        cv2.rectangle(test_img,(x,y),(x+w,y+h),(255,0,0),thickness=7)
        roi_gray=gray_img[y:y+w,x:x+h]#cropping region of interest i.e. face area from  image
        roi_gray=cv2.resize(roi_gray,(50,50))
        img_pixels = image.img_to_array(roi_gray)
        img_pixels = np.expand_dims(img_pixels, axis = 0)
        img_pixels /= 255

        predictions = model.predict(img_pixels)

        #find max indexed array
        max_index = np.argmax(predictions[0])

        #emotions = ('restless', 'doubt', 'attending')
        emotions = ('with helmet','without helmet')
        predicted_emotion = emotions[max_index]
        var=predicted_emotion

        cv2.putText(test_img, predicted_emotion, (int(x), int(y)), cv2.FONT_HERSHEY_SIMPLEX, 1, (0,0,255), 2)

    resized_img = cv2.resize(test_img, (1000, 1000))
    cv2.imshow('helmet analysis ',resized_img)
```

```
        if cv2.waitKey(10) == ord('q'):#wait until 'q' key is pressed

            break

    cap.release()
    cv2.destroyAllWindows()
    if(var=='without helmet'):
      d()
    else:
      warning()




import cv2
import numpy as np

def b():



    cap = cv2.VideoCapture(0)
    face_cascade = cv2.CascadeClassifier("haarcascade_frontalface_default.xml")


    cv2.namedWindow("Frame")
    while True:
        _, frame = cap.read()
        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
        luv= cv2.cvtColor(frame,cv2.COLOR_BGR2LUV)
        neighbours = 3
```

```
        cv2.imshow("Frame", frame)


        key = cv2.waitKey(1)
        if key == 67:
            cv2.imwrite("frame2.png",frame)
        it key == 99:

            break

    cap.release()
    cv2.destroyAllWindows()



def insert():
    #Create an instance of Tkinter frame
   global entry
   string= entry.get()
   label.configure(text="PLEASE INSERT YOUR CARD")

    #Initialize a Label to display the User Input
label=Label(win, text="", font=("Courier 22 bold"))
label.configure(text="PLEASE INSERT YOUR CARD")
label.pack()

#Create a Button to validate Entry Widget
Button(win,
           text="okay",
           command= a).pack()
win.mainloop()

btn.pack(pady=10)

# mainloop, runs infinitely
mainloop()
```

```python
from tkinter import *
from tkinter import ttk
import os
import cv2
import numpy as np
from tensorflow.keras.models import model_from_json
from tensorflow.keras.preprocessing import image


#Create an instance of Tkinter frame
win= Tk()

#Set the geometry of Tkinter frame
win.geometry("750x250")

def amount():
    #Create an instance of Tkinter frame
    win= Tk()


    #Set the geometry of Tkinter frame
    win.geometry("750x250")

    def display_text():
       global entry
       string= entry.get()
       label.configure(text="ENTER THE AMOUNT")

    #Initialize a Label to display the User Input
    label=Label(win, text="", font=("Courier 22 bold"))
    label.configure(text="ENTER THE AMOUNT")
    label.pack()
```

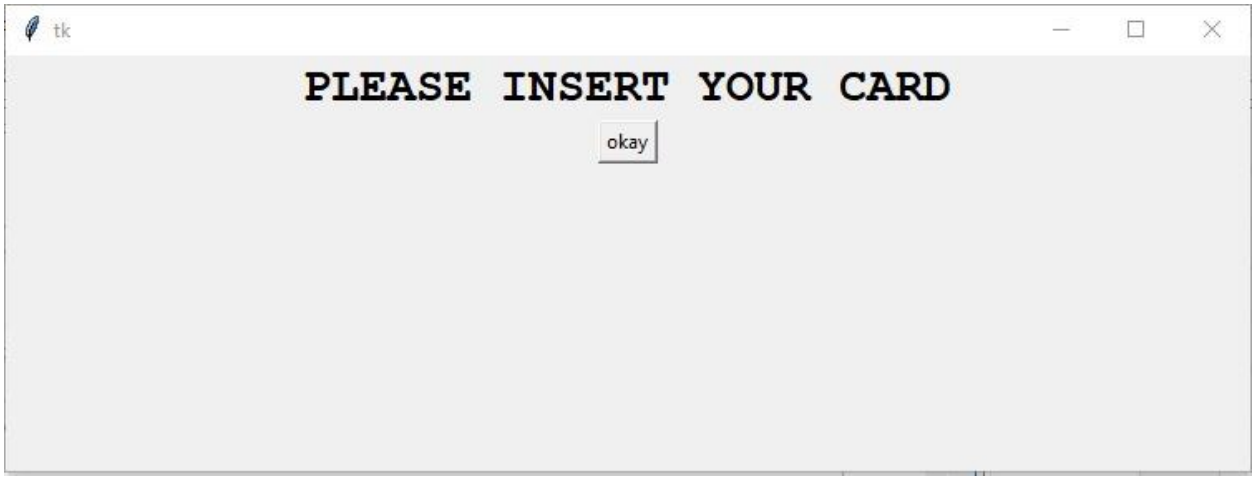
```python
    #Create an Entry widget to accept User Input
    entry= Entry(win, width= 40)
    entry.focus_set()
    entry.pack()

    #Create a Button to validate Entry Widget
    ttk.Button(win, text= "Okay",width= 20, command= transaction).pack(pady=20)

    win.mainloop()
def mask():
    i,j,k=0,0,0

    #load model
    model = model_from_json(open("modelmask.json", "r").read())
    #load weights
    model.load_weights('maskmodel.h5')


    face_haar_cascade = cv2.CascadeClassifier('haarcascade_frontalface_default.xml')


    cap=cv2.VideoCapture(0)

    while True:
        ret,test_img=cap.read()# captures frame and returns boolean value and captured image
        if not ret:
            continue
        gray_img= cv2.cvtColor(test_img, cv2.COLOR_BGR2GRAY)

        faces_detected = face_haar_cascade.detectMultiScale(gray_img, 1.32, 5)
```

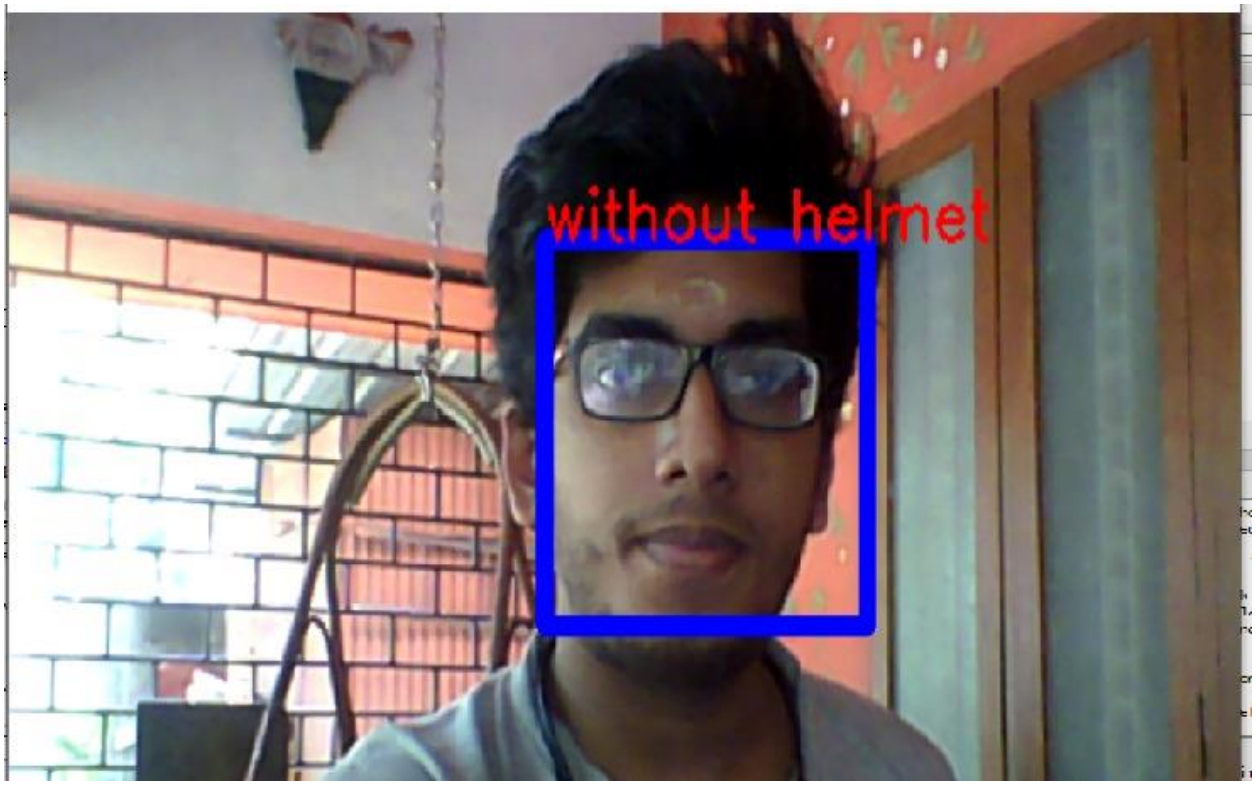
```
    for (x,y,w,h) in faces_detected:
        cv2.rectangle(test_img,(x,y),(x+w,y+h),(255,0,0),thickness=7)
        roi_gray=gray_img[y:y+w,x:x+h]#cropping region of interest i.e. face area from  image
        roi_gray=cv2.resize(roi_gray,(50,50))
        img_pixels = image.img_to_array(roi_gray)
        img_pixels = np.expand_dims(img_pixels, axis = 0)
        img_pixels /= 255

        predictions = model.predict(img_pixels)

        #find max indexed array
        max_index = np.argmax(predictions[0])

        #emotions = ('restless', 'doubt', 'attending')
        emotions = ('with mask', 'without mask')
        predicted_emotion = emotions[max_index]
        var=predicted_emotion

        cv2.putText(test_img,predicted_emotion , (int(x), int(y)), cv2.FONT_HERSHEY_SIMPLEX, 1, (0,0,255), 2)

    resized_img = cv2.resize(test_img, (1000, 700))
    cv2.imshow('Facial emotion analysis ',resized_img)



    if cv2.waitKey(10) == ord('q'):#wait until 'q' key is pressed

        break

cap.release()
cv2.destroyAllWindows()
if(var=='with mask'):
    warning2()
```

```
def pin():
   global entry
   string= entry.get()
   if(string=="1234"):
       amount()
   else:
       mask()
#Initialize a Label to display the User Input
label=Label(win, text="", font=("Courier 22 bold"))
label.configure(text="ENTER YOUR PIN")
label.pack()

#Create an Entry widget to accept User Input
entry= Entry(win, width= 40)
entry.focus_set()
entry.pack()

#Create a Button to validate Entry Widget
ttk.Button(win, text= "Okay",width= 20, command= pin).pack(pady=20)

win.mainloop()
def warning2():
    #Create an instance of Tkinter frame
    win= Tk()

    #Set the geometry of Tkinter frame
    win.geometry("750x250")

    def display_text():
       global entry
       string= entry.get()
       label.configure(text="PLEASE REMOVE YOUR MASK")
```

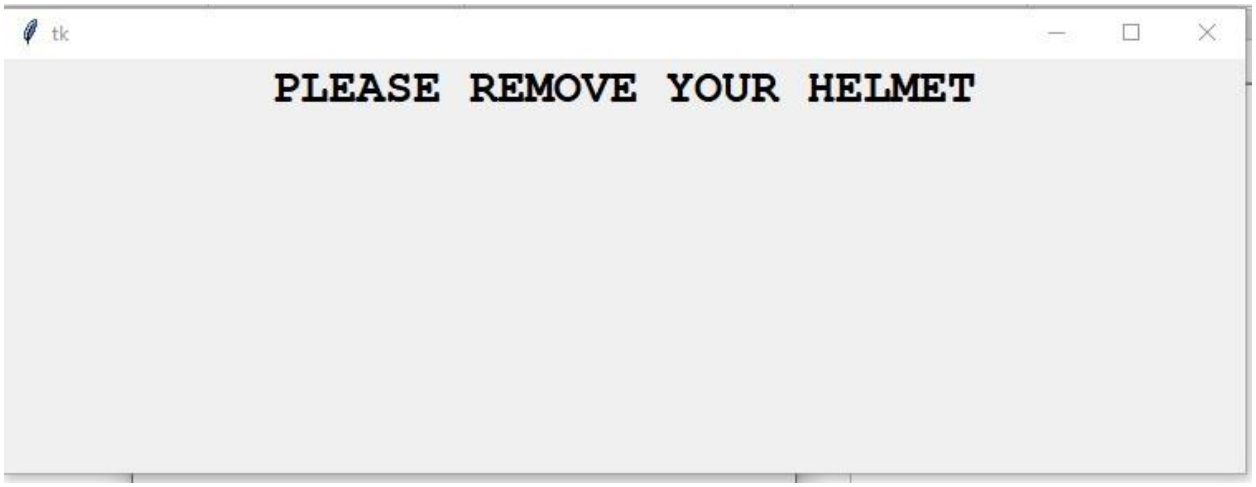
```
    #Initialize a Label to display the User Input
    label=Label(win, text="", font=("Courier 22 bold"))
    label.configure(text="PLEASE REMOVE YOUR MASK")
    label.pack()

#Create a Button to validate Entry Widget
    ttk.Button(win, text= "Okay",width= 20, command= pin).pack(pady=20)

    #Create a Button to validate Entry Widget

    win.mainloop()

def transaction():
    #Create an instance of Tkinter frame
    win= Tk()

    #Set the geometry of Tkinter frame
    win.geometry("750x250")

    def display_text():
       global entry
       string= entry.get()
       label.configure(text="TRANSACTION SCUCSSFULL")

    #Initialize a Label to display the User Input
    label=Label(win, text="", font=("Courier 22 bold"))
    label.configure(text="TRANSACTION SCUCSSFULL")
    label.pack()

#Create a Button to validate Entry Widget

    #Create a Button to validate Entry Widget
    win.mainloop()
```

13.2 Screenshots

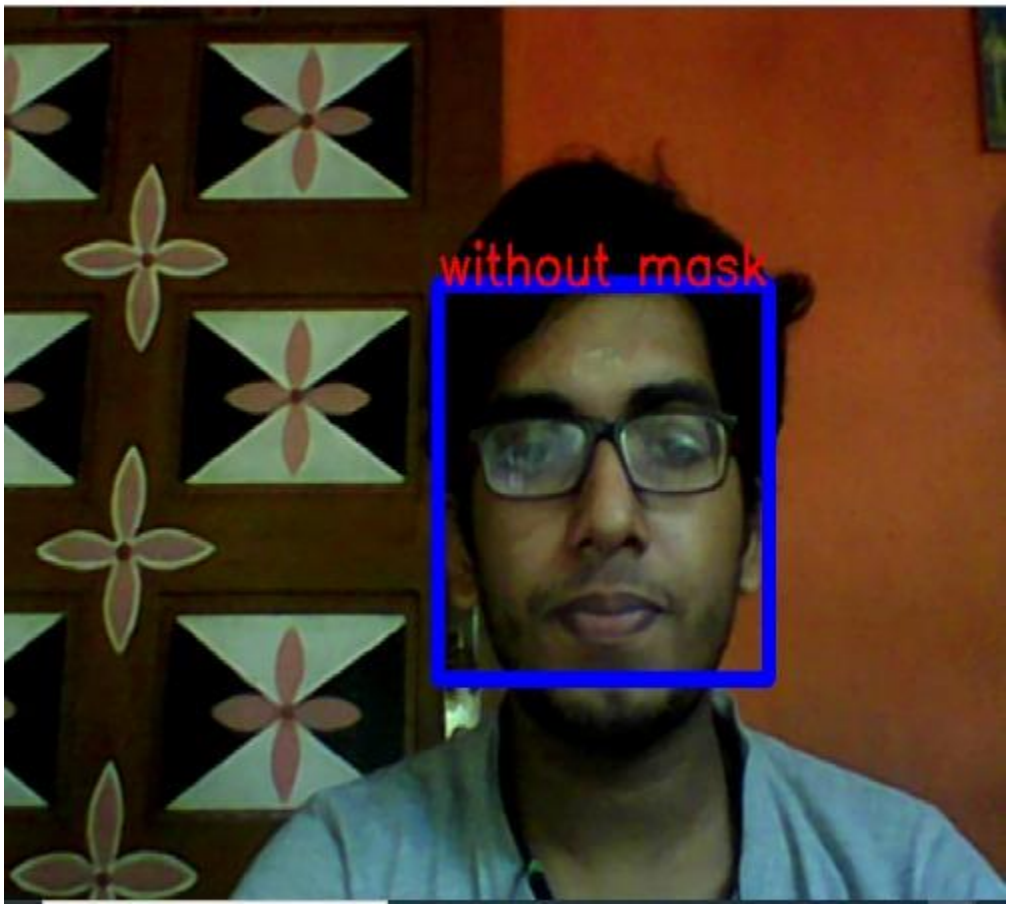




with helmet

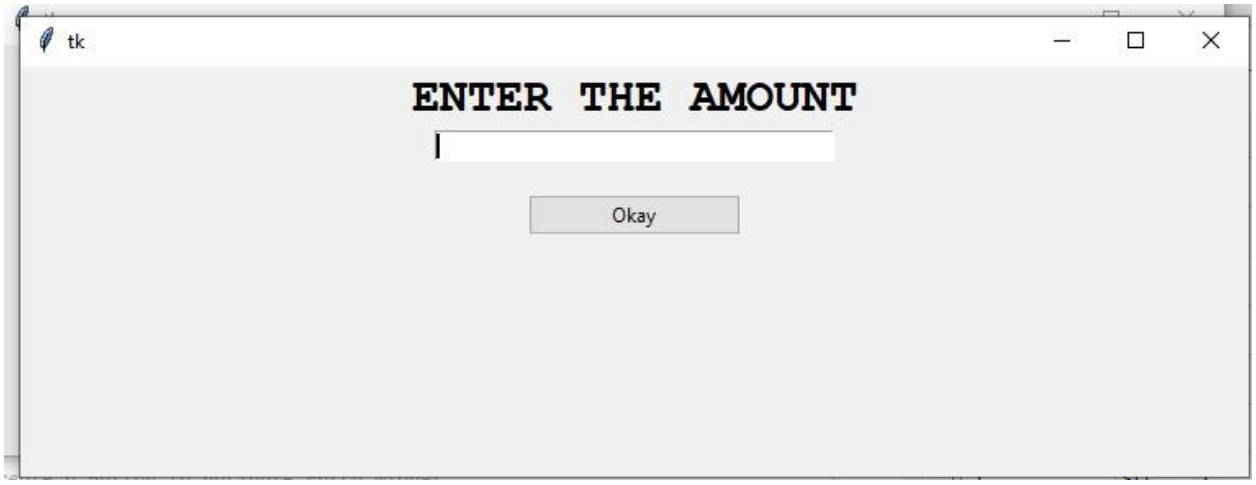

tk
PLEASE REMOVE YOUR HELMET

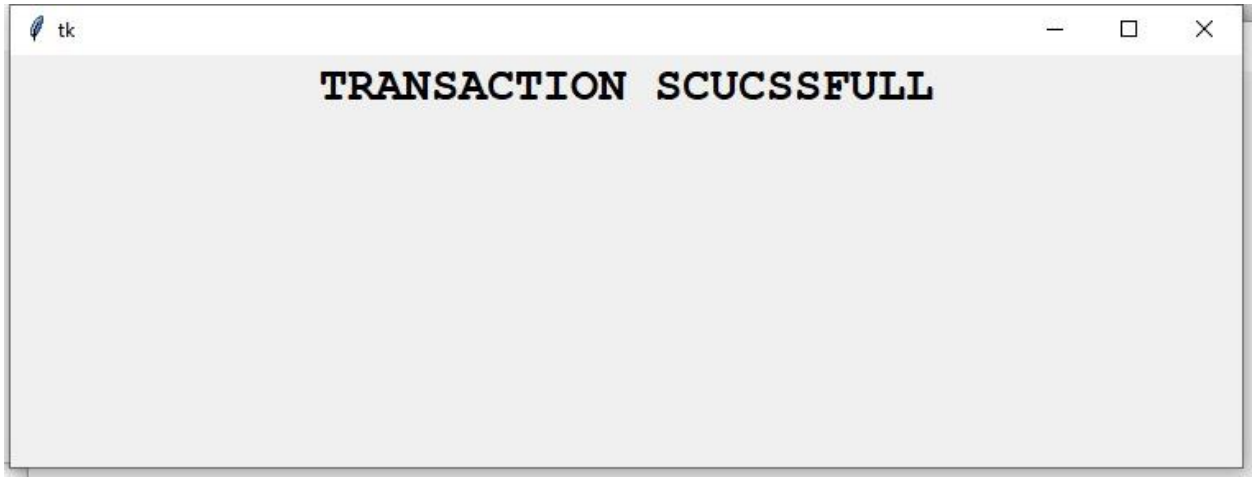
tk
ENTER YOUR PIN
1234
Okay


without mask

with mask

tk
PLEASE REMOVE YOUR MASK
Okay


tk
ENTER THE AMOUNT
Okay


tk
TRANSACTION SCUCSSFULL